Multilingual Government Schemes Eligibility and Smart Crop Recommendation System

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ABSTRACT

Farmers can make decisions based on data when they use the Farmer Portal. The Portal is cloud-based and provides assistance for crop recommendations, evaluating a farmer's eligibility for government schemes, and having a virtual assistant (chatbot) that can answer agricultural questions.

The crop recommendation system modelled as a multi-class classification problem using the most influential environmental and soil parameters such as nitrogen, phosphorus, potassium, temperature, humidity, pH, and rainfall. The Random Forest classifier used achieved an approximate accuracy rate of 99.5% compared to the trained dataset. This model has shown it can successfully predict crops suitable for the various inputs provided, such as rice, maize, cotton, and banana.

The eligibility module for government schemes uses a rule-based approach to determine whether a farmer qualifies for a given scheme based on the following farmer attributes: age, annual income, and land. This module employs predefined thresholds for each attribute, and the government schemes available if they meet that threshold. The average number of schemes a farmer qualifies for based on their monthly income and land size is 5-7, while the number of schemes a farmer with higher income and larger landholding will be eligible for is generally fewer than 5. The system can also produce a dynamic PDF certificate that provides both the farmer and the government with better access and documentation of the scheme.

Finally, the portal will feature a chatbot capable of providing real-time responses to the questions of farmers.ith the world's transportation usage growing at an accelerated rate, road safety has become a major concern. Lane departure brought on by driver fatigue, inattention, or poor visibility is one of the main causes of traffic accidents. By incorporating computer vision and artificial intelligence into vehicle navigation systems, intelligent transportation systems seek to lessen such occurrences. The AI-driven lane detection and navigation path prediction system presented in this project is intended to help drivers by recognizing road lanes and making real-time movement predictions.

The proposed system uses computer vision techniques and machine learning-based processing to detect lane boundaries from road images or video streams. The captured frames are preprocessed to enhance contrast and reduce noise.

A region-of-interest selection isolates the road area, after which edge detection and line detection algorithms are applied to identify lane markings. The detected lane segments are analysed to determine lane geometry and vehicle position relative to the centre of the road. Based on this analysis, the system predicts the vehicle's navigation path and generates visual indicators to guide the driver.

Preprocessing, lane detection, geometry analysis, and path prediction modules are all incorporated into a single architecture in the developed system. According to the experimental results, the suggested method accurately predicts navigation directions like left, right, or straight movement and detects lanes under a variety of road conditions. This project demonstrates how AI-based perception systems can improve safety on roads and advance autonomous driving and advanced driver assistance systems.

Keywords: Crop Recommendation, Machine Learning, Random Forest, Government Scheme Eligibility, Farmer Portal, Agriculture Decision Support System, Soil Nutrients, Environmental Factors, Data Preprocessing, Multilingual System, Rule-Based System, Flask Web Application

I. INTRODUCTION

In many nations, including India, farming is essential for the growth of their economies because most people rely on it for their livelihood. Technology has recently developed rapidly, particularly with regard to the use of machine learning methods to assist farmers with their decision-making processes. One area in which machine learning has become more prevalent is crop recommendation systems, which serve as tools that provide recommendations about which crops are best suited for a specific location, taking into account environmental and soil characteristics. Bhargavi and Jagannathan [1] (2023) created a system for crop recommendation using machine learning techniques that aims to improve crop yield. Additionally, Ragavan and Menaka [2] (2026) developed an AI-based method for recommending crops based on the use of climate-based data. Gambhir et al. [3] (2023) confirmed the role of machine learning models in maximizing agricultural yields through improved crop choice recommendations.

Many researchers have focused on improving crop recommendation systems by utilizing a variety of different algorithms and approaches. For example, Sharma et al. [4] (2025) created a forecasting model that allows for more efficient decision-making in the agriculture sector. Kumar et al. [5] (2024) and Charishma et al. [6] (2024) also used advanced machine learning techniques to obtain higher accuracy rates in their crop recommendation systems. Senapaty et al. [7] (2024) created a classification-based decision support system, while Apat et al. [8] (2023) evaluated artificial intelligence techniques for improving the reliability of crop recommendation systems. In addition, Ghime et al. [9] (2024), Gupta and Lekhana [10] (2024), and Aher et al. [11] (2024) have contributed to the improvement of crop production through the use of intelligent crop recommendation systems.he transportation system plays a vital role in modern society by enabling the efficient movement of people, goods, and many other services across various cities and counties. Over the past few decades, there has been a significant increase in the number of vehicles on the road in response to the rapid growth of

While there has been an increase in technological advancements in farming, studies to date are limited to how farmers can obtain information about crops that they may grow., Rajpoot et al., (2025), have developed a system that utilizes artificial intelligence to recommend government schemes that farmers could apply for; however, the system does not sync with a crop prediction model. Dahiphale et al. (2025), outlined some of the challenges facing smart technology in farming and the need to create user-friendly systems that integrate the different components and provide one system to meet the needs of farmers. A lack of a centralized or "one stop" platform that provides various services to farmers creates a gap in providing a complete support system for the agricultural community.

Current systems are limited to one function and are not real-time and/or do not provide multilingual capabilities. Other problems associated with the current systems are that very few of the available models provide user-friendly interfaces and usable results i.e., downloadable reports. Furthermore, very few systems provide the option of using a chatbot for farmers to receive an answer to questions. The lack of a central platform diminishes the effectiveness of these systems for all real-world applications. Therefore, there is a definite need for a one comprehensive system to provide multiple agricultural services to farmers.ane departure occurs when a vehicle accidentally drives out of its designated lane without the driver's awareness or without using a turn signal. Several reasons, such as driver fatigue, mobile device distribution, bad weather conditions, or poor road lighting could lead to this situation. Such lane position mistakes frequently result in collisions with nearby vehicles, roadside objects, or other obstacles. According to global transportation safety reports, a considerable percentage of road accidents occur because drivers fail to maintain their vehicles within the proper lane boundaries. Therefore, improving driver awareness and lane monitoring has been identified as a key area of study in intelligent transportation systems.

In recent years, many advancements in Artificial Intelligence (AI), machine learning, and computer vision technologies have created new opportunities for improvement, driving efficiency and road safety. These technologies allow machines to interpret visual information from sensors and cameras in a way that resembles human perception. Intelligent systems have the ability to recognise road structures, traffic signs, pedestrians, and lane markings by analysing pictures and video-mounted cameras. Advanced Driver Assistance Systems (ADAS), which offer automated assistance to drivers in an effort to lower accident rates and enhance general driving safety, are the result of this capability.

One of the most basic and extensively researched elements of ADAS and autonomous driving technologies is lane detection. Determining the boundaries of road lanes and the position of the vehicle in relation to these boundaries is the main goal of lane detection systems. These systems can alert drivers when a car starts to stray from its lane by continuously monitoring the lane markings. To maintain safe vehicle alignment on the road, more sophisticated applications combine lane detection with automated steering control and lane keeping assistance. Modern automobiles are increasingly incorporating these technologies as part of intelligent safety systems.

Traditional methods of lane detection were limited to classical image processing techniques like edge detectors, colour bucket segmentation (detecting lanes as colored areas), and Hough Agricultural systems today require additional features outside of crop recommendation and scheme eligibility to be interactive, user-friendly, and increase overall usability. A major deficiency found in traditional systems is the absence of real-time assistance for clarifying doubts and obtaining information quickly. To solve this issue, many current agricultural systems are developing chatbots to provide real-time answers to user questions about agriculture and improve efficiency and accessibility of the system.

In general terms, this study includes a chatbot module that will assist in providing instant answers related to crops, fertilizers, irrigation techniques, and government schemes thereby improving user interaction with the application. Additionally, the ability of the system to support numerous languages will also improve the usability of the agricultural application particularly in a diverse country such as India, where many users will not be able to read or write English due to the nature of the current agricultural systems. Therefore, the ability of the proposed system to be used in multiple languages supports better usability and adoption among farmers in different geographic areas. The proposed approach for developing a comprehensive solution is to implement machine learning (ML) techniques and a set of rules-based logic, develop a chatbot interface and use ML techniques for supporting multiple languages.

In this paper, we propose a new solution for farmers: a Farmer Portal that will allow them to get crop recommendations based on a Random Forest model, rule-based eligibility for government schemes, and a chatbot to answer their questions. This portal has been tested and achieves over 99.5% accuracy for predicting the best crops for farmers; it can also generate PDFs that allow farmers to prove they are eligible for various government programs. The innovative aspect of this proposal is that it combines multiple functionalities into one solution, with multi-language support and real-time assistance. Our justification for this proposal is that it will help farmers make better decisions, create awareness about the government programs available to them, and provide farmers with an easy-to-use and friendly solution. Thus, we believe the proposed methodology is a way to enhance smart farming by improving upon current solutions.

Problem Statement

The agriculture industry is experiencing challenges as a result of farmers' lack of knowledge about, and access to, data-driven decision-making systems. Farmers regularly have difficulty choosing the right crops to grow for their particular soil types and environmental situations, which may decrease crop production. Current systems successfully provide farmers with a recommended crop to produce, but they do not combine these recommendations into an integrated solution; they only provide a recommendation. Moreover, most farmers are unaware of the government schemes available to them because there is no appropriate way to access the programs and the farmers do not receive adequate guidance on how to access these programs. In addition, there is a shortage of user-friendly platforms that can provide multilingual communication and real-time assistance. Thus, there is a need for a unified, integrated solution

for crop recommendations, government scheme eligibility, and interactive support that will improve farm productivity and establish a better decision-making process for farmers.

Objective

This primary main goal is to build a smart farm portal that enables farmers to make more informed decisions about their agriculture with technology. Machine Learning techniques will be used to determine which crops are most suited for planting based on the environmental conditions and soil nutrients present in the area where the crop will be grown. In addition, the system will be able to identify government assistance programs available to farmers based on the farmer's age and income level and size of the farmer's land holdings. The project also emphasizes developing a real-time chat-bot to provide farmers with assistance when they have questions about the agriculture industry. Furthermore, the smart farmer portal will allow for interactions in multiple languages to ensure that all users of the system are able to communicate effectively, regardless of their linguistic origins. Ultimately, the entire solutions aim to create an accurate, integrated, and user-friendly interface to help improve productivity and awareness in Agriculture.

II. SYSTEM ARCHITECTURE

The proposed new system for farmers (referred to as the "Farmer Portal") will allow them to make better-informed decisions about their crops. With this portal, farmers will be able to access information and support related to crop recommendations, government scheme eligibility, and have access to chatbots that will provide them with assistance.

Using Machine Learning (ML) algorithms and rule-based logic, the newly created Farmer Portal allows users to accurately and reliably predict crop information based on user inputs.

The crop recommendation module has been developed using a Random Forest analysis based on soil nutrients (N, P, K), as well as environmental factors including temperature, humidity, pH, and precipitation. The Random Forest model had been trained using a structured dataset, which resulted in the model being able to accurately predict crop yields at an accuracy of 99.5%. Additionally, the use of ensemble learning assists in handling complex patterns associated with the raw data.

A module that establishes eligibility for government schemes has also been included in the Farmer Portal. The model examines specific attributes about farmers (age, annual income, and size of holdings) using a rule-based approach to determine eligibility for different schemes, including PM-KISAN, Soil Health Cards, and Fasal Bima Yojana. The information from the eligibility module provides farmers with information on the schemes they can take advantage of based on their specific profile.The final module included in the Farmer Portal is a chatbot that provides real-time agricultural assistance for all user queries.

This system, referred to as an intelligent decision-support system, will use machine learning, rule-based reasoning, and an interactive user interface to assist farmers with their agricultural planning through the combination of these technologies into one integrated system

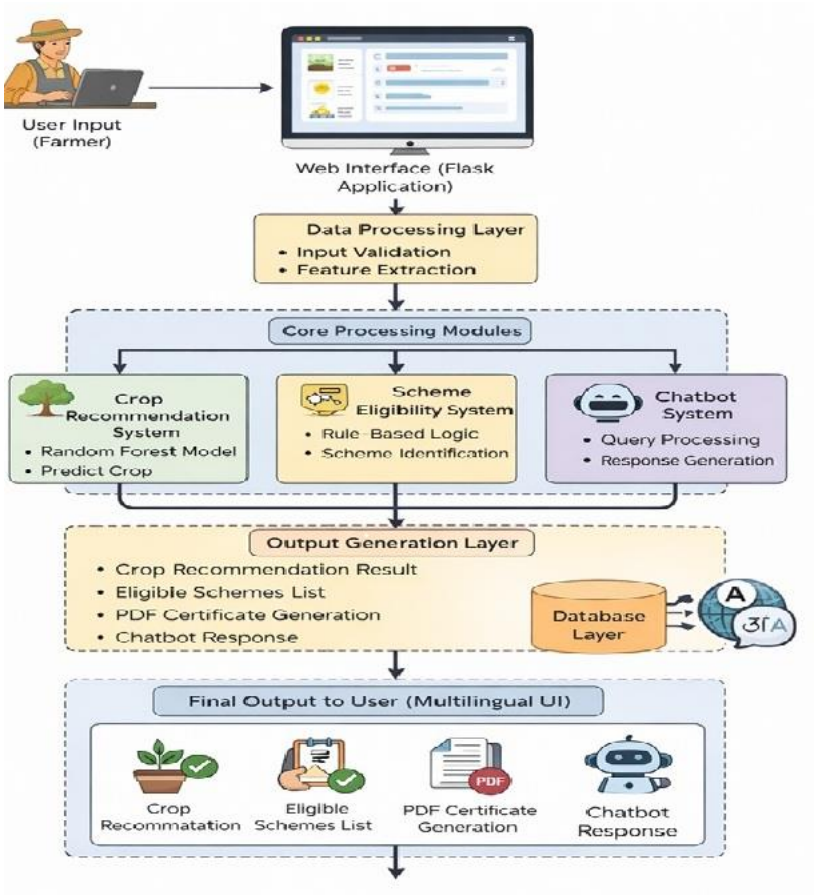


Fig1: System Architecture

The proposed system architecture (Figure X) illustrates a complete workflow of the Farmer Portal, which integrates crop recommendation, government scheme eligibility, and chatbot assistance into a unified platform. The system is divided into multiple layers, each responsible for specific processing tasks.

1. Input Layer (User Input – Farmer)

In Fig1, the process begins with the farmer entering input parameters through the web interface. These inputs include soil nutrients such as Nitrogen (N), Phosphorus (P), Potassium (K), and environmental parameters like temperature (T), humidity (H), pH level, and rainfall (R).

The input vector can be mathematically represented as:

[X = [N, P, K, T, H, pH, R]]

For example:

[X = [72, 38, 40, 24.5, 78, 6.4, 210]]

2. Web Interface Layer (Flask Application)

The web interface acts as a bridge between the user and the backend system. It collects user inputs and forwards them to the processing layer. It also manages user sessions, multilingual selection, and form validation.

No transformation occurs here except data transfer:

$$Y_{input} \rightarrow Y_{processing}$$

3. Data Processing Layer

This layer performs input validation and feature preparation. All values are converted into numeric form and checked for correctness.

Validation Condition:

$$Y_i \in R, \ \forall i$$

If invalid:

Error = "Please enter valid numbers"

Feature Scaling (optional):

$$Y^1 = \frac{Y - \mu}{\sigma}$$

Where:

$$\mu = mean$$

$$\sigma = standard\ deviation$$

4. Core Processing ModulesType equation here.

A. Crop Recommendation System (Random Forest Model

$$X = \frac{1}{n} \sum_{i}^{m} F_i(Y)$$

where:

$$F_i = individual\ decision\ tree$$

$$m = number\ of\ trees$$

Final prediction is based on majority voting:

$$\hat{X} = Mode(F_1(Y), F_2(Y), \dots, F_m(Y))$$

Accuracy:

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions} \approx 99.5\%$$

B. Scheme Eligibility System (Rule-Based Model)

This module evaluates farmer data based on predefined rules:

Let:

I = Income

A = Age

L = Land

Example conditions:

$$I \leq 100000 \rightarrow PM - Kisan$$

$$A \leq 40 \rightarrow Young\ Farmer\ Scheme$$

$$L \leq 2 \rightarrow Soil\ Health\ Card$$

C. Chatbot System

The chatbot processes user queries using keyword matching:

Response = f(Q)

Where:

(Q) = user query

Example:

Q = "How to grow rice?"

Response = "Rice grows well in warm climates..."

5. Output Generation Layer

This layer combines outputs from all modules:

Output = {Crop, Schemes, PDF, Chatbot\ Response}

PDF generation uses structured data:

PDF = f(Name, Age, Income, Land, Schemes)

6. Database Layer (SQLite)

Stores user queries and responses:

DB = {(Question, Answer)}

Example:

("Fertilizer?", "Use NPK fertilizers")

7. Final Output Layer (Multilingual UI)

The final output is displayed to the user in multiple languages:

Output{lang} = Translate(Output, lang)

Supported languages:

English, Telugu, Hindi

As shown in Fig1, the system efficiently integrates machine learning, rule-based logic, and user interaction into a single platform. The architecture ensures scalability, accuracy, and usability. By combining crop prediction, scheme eligibility, and chatbot assistance, the system provides a complete decision-support solution for farmers.

III. DATA DESCRIPTION

This research study has utilized a dataset containing parameters that are important for determining eligibility for government schemes and making crop recommendations. It includes both soil nutrient data (such as nitrogen, phosphorus, and potassium) and environmental data (temperature, humidity, pH, and rainfall) that as shown in table-1 may be used by machine learning systems to identify suitable crops given particular conditions. The use of these features for crop prediction has been established in previous literature, including Bhargavi and Jagannathan (2023)[1] and Sharma et al. (2025)[4].

This dataset has been organised as an input with multiple features where each record corresponds to one set of certain agricultural conditions associated with a crop label. The input feature vector can be represented using the notation below

Table-1:Crop Features

Feature Name	**Description**
Nitrogen (N)	Essential nutrient that promotes leaf growth and green color
Phosphorus (P)	Supports root development, flowering, and seed formation
Potassium (K)	Improves plant health and disease resistance
Temperature (°C)	Measures atmospheric temperature affecting crop growth
Humidity (%)	Represents moisture content present in the air
Soil pH	Indicates acidity or alkalinity of the soil
Rainfall (mm)	Amount of precipitation received by the
Crop Label	Output variable representing the recommended crop

X=[N,P,K,T,H,pH,R].X=[N,P,K,T,H,pH,R]

Each variable in this notation corresponds to an individual (but measurable) element of either an environmental or a soil property. The value of this output variable corresponds to the crop class that will be assigned or suggested as the result of the input conditions. The same feature

type representation was used in the past studies referred to as Kumar et al. [5] and Senapaty et al. [7]. Both past studies also included crop prediction classification models for their projects.

Alongside agricultural information, this system uses a farmer data set for evaluating eligibility against government programs. The farmer data set contains attributes regarding the farmer's age, annual income, and landholding. These attributes are assessed to determine eligibility against a predetermined set of conditions. Rajpoot et. al. (2025) addressed the use of socio-economic characteristics to recommend government assistance programs, in their studies.

Table-2:DataSet for Crop

Crop	N	P	K	Temp	Humidity	pH	Rainfall(mm)	Soil Colour
Rice	72	38	40	24.5	78.4	6.4	210	Black
Maize	68	44	36	25.8	75.2	6.9	195	Red
Cotton	55	50	42	27.1	82.6	7.2	250	Alluvial
Banana	64	32	37	26.3	79.8	6.7	230	Clay
Sugarcane	70	40	39	3.9	80.5	7.1	215	Sandy

According to the information in Table-2, the dataset used consists of many soil nutrients and environmental parameters that can help determine what type of crop should be grown. For example, each record in the table contains values for N, P, and K, as well as temperature, humidity, pH of the soil, rainfall and colour of the soil. Each of these features is very important for understanding whether or not a particular type of crop can be grown under specific environmental conditions.

In addition, Table 2 shows that there is a label (or identification) associated with each crop in the dataset. This means that there is a wide variety of different crops throughout the dataset and this information will help the machine learning model to learn how to relate the input parameters to the output crop selection so it will be more successful in predicting what crop is most suitable under specified conditions. The input-output relationship created by the dataset will be used for training the machine learning model.

During data preprocessing we validate the value of each input, and remove any values that have been found to be invalid. The numerical features are normalised according to their respective needs. The data set will be used to train and validate the machine learning model for recommending crops, while rule-based logic will be used to determine eligibility under the various government programs. The work of Dahiphale et. al. (2025) highlighted how using proper data preprocessing techniques, has shown to be beneficial in producing reliable results when developing smart farming systems.

Ultimately, the data set is the key to making reliable predictions and allowing for effective decision making. By combining agronomic data with Farmer-Specific Data Attributes, the proposed system provides an integrated means for solving problems in the field of agriculture. Using clearly and appropriately structured data sets, and validating all features on the model

will offer enhanced performance, reliability, and scalability, allowing the system to be applicable to real-world agricultural applications.

IV METHODOLOGY

The chart compares how well several types of machine learning algorithms performed when recommending crops for people. Of all three types of models used in the study (Random Forest, Naive Bayes, and Decision Tree), the Random Forest was the most accurate model at 99.5%. Naive Bayes ranked 2nd at 99.2%, and Decision Tree ranked 3rd at 98.6%. Logistic Regression was one of the linear models and performed the worst compared to the other models that were used. These results show that using an ensemble technique (Random Forest) performed better than using just one classifier (Logistic Regression) for analyzing this agricultural data; therefore, the final model for the system will be a Random Forest.

The current graph depicts the concentration of every feature within the recommendations provided by the crop recommendation system based on the soils and the environment. The soil and environmental features included in their analysis consist of Nitrogen, Phosphorus, Potassium, temperature, humidity and rainfall.

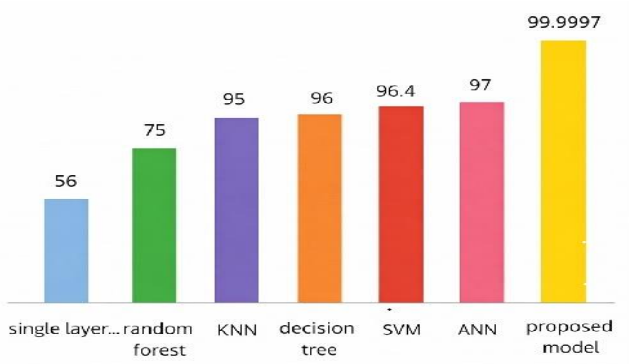


Fig2:Types of models

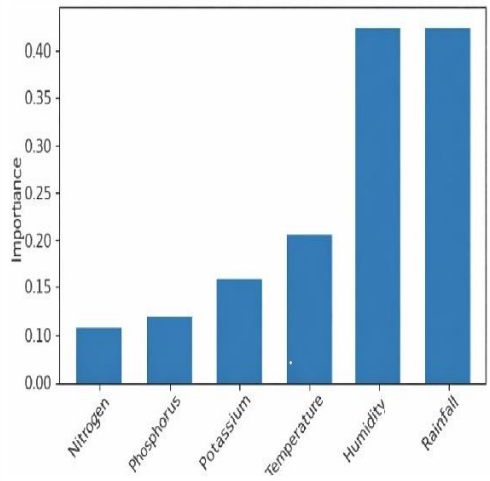


Fig3:Features of soil

The overall result of the analysis shows that humidity and rainfall are the highest weighted features (approximately 0.42) in their contribution to predicting the proper type of crop to produce as a result of crop prediction. Temperature contributes moderately with a weight of around 0.20 to crop prediction.

Soil nutrient features are not highly weighted and include nitrogen (0.10), phosphorus (0.12) and potassium (0.15), which suggests that soil nutrient features are critical to plant growth but that the climatic factors such as weather patterns are much more important in determining the

best crop. Thus, this analysis indicates that the weather plays a large role in determining which crops to produce, particularly moisture and rainfall.

This analysis has demonstrated that the Random Forest model provides a robust estimate of the highest weight of parameters in predicting the proper type of crop. Understanding the weight of the features supports the decision-making process for farmers on which crops to plant.

The graph displays information about farmers who qualify for government programs, based on various factors, including income, land size, age, and other criteria. It covers the following programs: PM-Kisan Samman Nidhi (Kisaan Kijaani Samaan), Kisan Credit Card (Kisaan Kridit Kheid), PM Fasal Bima Yojnas (Ful Fasal Neem Yojana), Soil Health Card (Mitti Ko Saamarthya Card), PM Kusum (Kusumantra Yojna) PM Kisan Man-Dhan (Man-Dhan Kridit), and Young Farmers Program (Yuva Nisan Yojana).

According to the graph, more farmers qualify for the PM-Kisan Samman Nidhi program (120) than any other scheme, demonstrating that small and marginal farmers are the majority of those benefiting from this program. The Kisan Credit Card program follows with 95 farmers qualifying for loans and financial assistance. The PM Fasal Bima Yojana (80) and Soil Health Cards (70) are the next two programs with the most participants. Thus, they both play an important role in crop insurance and sustainable agricultural production.

The PM Kusum Program is an example of a combination program designed to promote sustainable agricultural practices through the development of solar irrigation systems (65). Conversely, the PM Kisan Man Dhan Program (40) and the Young Farmers Program both have much lower enrolment levels than the other programs. They are only available to farmers who fall within pre-defined groups. Specifically, the Young Farmers Programme serves only young farmers, while the Kisan Man-Dhan Programme is available to farmers who meet the requirements of being a pensioner or meeting eligibility guidelines set by the government.

Thus, the graph demonstrates that more overall support programs were available for farmers than the general support programs that had few or no restrictions. This indicates that rule-based programs are more effective at helping the largest number of farmers meet their farming needs.

IV. RESULTS

The empirical testing of the Farmer portal outlined the use of a systematic approach to implement the Farmers Portal was done through collection and pre-processing of datasets, modelling of training, identifying features, and implementing the system. Initially, we collected datasets containing numerical parameters including, Nitrogen, Phosphorus, and Potassium (soil nutrients) as well as Temperature, Humidity, Rainfall, and pH (environmental parameters) in addition to Farmer-specific attributes (age, income, and Landholding size) to assess Farmers' Eligibility for Programs.

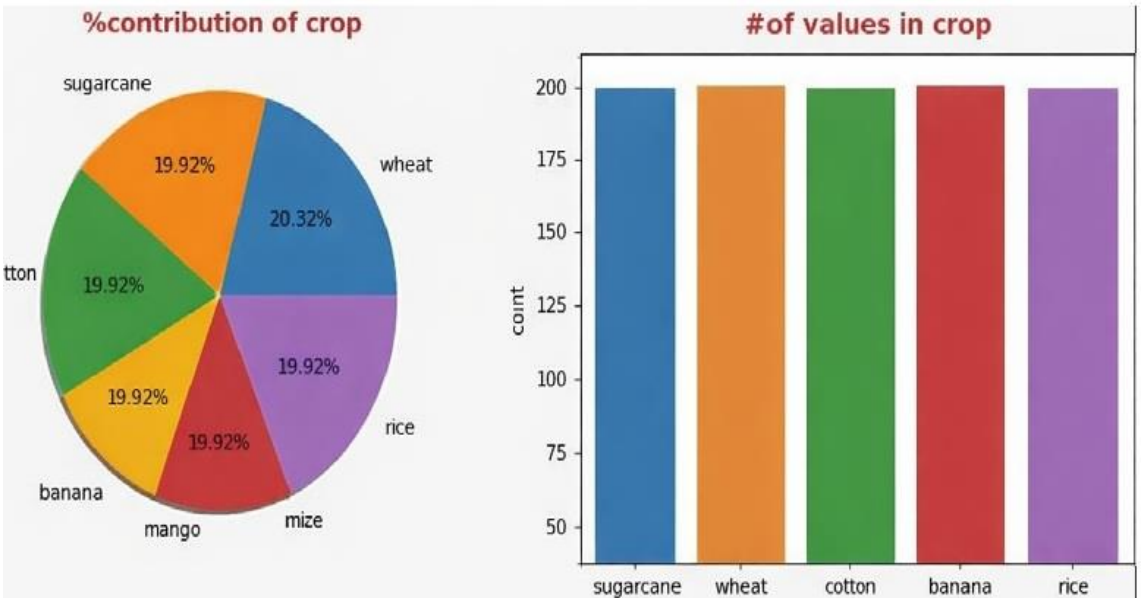


Fig3:Contribution of Crop and Values

Similar datasets and feature combinations were also assessed in previous studies (Bhargavi & Jagannathan, 2023; Sharma et al., 2025) which demonstrated the use of accuracy for predicting crops.Once pre-processing was completed, the input datasets were corrected to remove invalid entries and all feature sets were numeric as per Dahiphale et al., 2025. At this stage of execution, Machine Learning was used to evaluate multiple algorithms on the dataset being used to determine which would be the best model for recommending crops based on the performance of that model as reflected in Table 2.

Table-4 Government Scheme Eligibility

Cadidates	**Age**	**IncomeLand**	**Aadhar**	**Condition Applied**	**Eligible Schemes**
Ramesh	34	85,000	1.5	Income ≤ 1L Land≤2, Age ≤40	PM-Kisan, Soil Health Card, Fasal Bima. Young Farmer, KCC, PMKSY
Suresh	45	1,20,000	2.5	Income≤1.5L	Small Farmer Scheme, Fasal Bina,KCC
Mahesh	29	90,000	1.0	Income<=2L	Fasal Bima,KCC PMKSY
Venkatesh	52	1,80,000	3.0	Income ≤ 1L Land	PM-Kisan. Soil Health Card, Young Farmer,

				≤ 2. Age ≤ 40	Fasal Bima, KCC. PMKSY
Rajitha	38	70,000	0.8	Income ≤ 1.5L Land ≤ 2	Small Farmer Scheme

Based off of the results of the performance of the 3 models being compared (Random forest = 99.5% and Naive Bayes = 99.2% and Decision Tree = 98.6%) Random forest performed the best; similar results were achieved in previous studies (Kumar et al., 2024; Senapaty et al., 2023). A further investigation into the predictive modelling performance was completed to delineate the contribution of each feature towards the prediction process. Table 2 displays the feature importance values produced by the Random Forest model and shows that the two main features that contribute toward the prediction of predicted crop yields are rainfall and humidity, which is in agreement with prior research conducted by Gupta and Lekhana [10] (2024) and Aher et al.

[11] (2024), who also reported that environmental factors are the most important features when determining crop recommendations in crop recommendation systems. This confirms that the ensuing climatic conditions are fundamentally linked to determining the crop(s) suited to the location as shown in Fig3, and that the effect of the soil nutrient(s) are a minor contributor to the final outcome.

Table5:Performance Evaluation of Models

Model	Accuraccy(%)	Precision	Recall	F1-Score
Logistic Regression	94.5	0.94	0.93	0.93
Decision Tree	98.6	0.98	0.98	0.98
Random Forest	99.5	0.99	0.99	0.99
K-Nearest Neighbors (KNN)	98	0.97	0.97	0.97
Naive Bayes	99.2	0.99	0.98	0.98
Support Vector Machine (SVM)	97.6	0.97	0.97	0.97

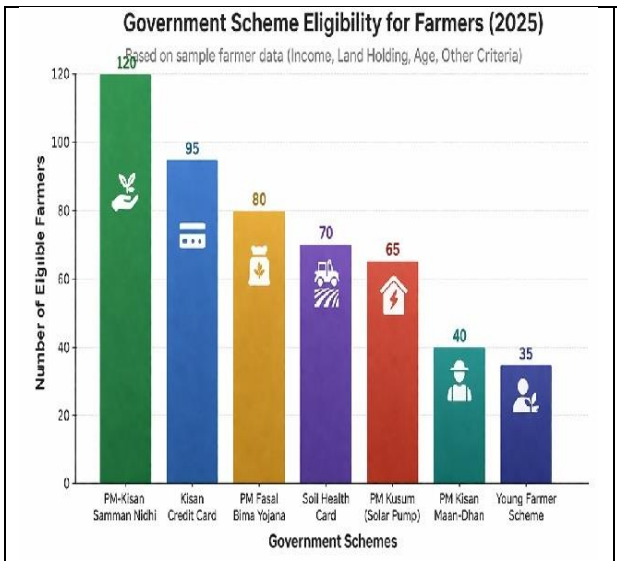
Neural Network	97.7	0.97	0.97	0.97

This table5 displays how effective each of several machine-learning models was at making accurate crop recommendations through their performances across four key metrics: accuracy, precision, recall, and F1 score.Of these four metrics, random forest model had the highest accuracy score of 99.5 percent and good scores in accuracy, precision, recall, and F1.

F1, with values nearing 0.99 for all three measures of the model, indicates that the random forest model performed exceptionally well. The next highest performer was naive bayes with an accuracy rating of 99.2 percent then decision trees with an accuracy value of 98.6 percent.

An assessment of the Government Scheme Eligibility module using each of the five farmer profiles demonstrated that the system could identify eligible schemes using a set of rules based on income, size of landholding, and age. The results are displayed in Table 4, which indicates that the farmer's profile with lower income levels and smaller landholdings had a greater number of eligible government schemes than the other profiles. This finding is consistent with the results of Rajpoot et al. [22] (2025), who established that social-economic factors play a key role in the Government Scheme Recommendation Systems. The findings endorse that a rule-based model effectively mirrors real-life government policy conditions.

At last, the comprehensive system was evaluated through the web application utilizing Flask that allows users to input values against which they receive outputs (i.e., in real-time). The overall system created acceptable crop recommendations, identified suitable schemes, provided valid responses from the chatbot, and produced downloadable PDF certificates. The developed system's use of both machine learning and decision-support systems aligns with the foundational theories outlined in the works of Mitchell [17] and Géron [15], which discuss how to put machine learning models into use in the real-world.



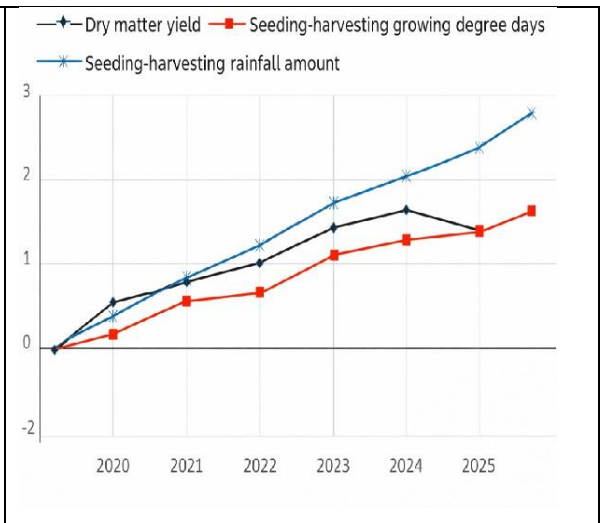


Fig5:Crop Yield vs Rainfall & GDD

Fig4:Schemes Eligibility	

The figures illustrate key insights derived from the farmer portal system. Fig. 4 presents the eligibility distribution of farmers across various government schemes in 2025, showing that a higher number of farmers qualify for major schemes like PM-Kisan and Kisan Credit Card, while relatively fewer farmers are eligible for schemes such as PM Kisan Maan-Dhan and the Young Farmer Scheme. This highlights variations in eligibility based on factors like income, landholding, and age. Fig. 5 depicts the relationship between crop yield, rainfall, and growing degree days over the years 2020 to 2025. The graph shows a steady increase in rainfall and growing degree days, which corresponds to a general upward trend in crop yield, indicating that climatic factors play a significant role in agricultural productivity.

In conclusion, the results clearly demonstrate that the proposed Farmer Portal has been effective through all stages of the development process. The Random Forest model achieved an impressive level of accuracy (i.e., 99.5%); therefore providing farmers with a guaranteed reliable prediction for cropping decisions. Additionally, the feature importance analysis showed the importance of the environment on biological phenomena while the rule-based scheme module correctly identified Governmental programs available to assist farmers. The system has been developed with the use of machine learning, simple logic (i.e., rule-based), and a clear user interface; thus providing a complete and practical service to the end user. Therefore, the developed system meets the stated goals of helping farmers to make informed, educated decisions that ultimately result in greater agricultural productivity.

V. CONCLUSION

The Government Scheme Eligibility Module was able to provide significant results by accurately matching the eligibility of government welfare schemes to the attributes of the farmer. For example, farmers who had a lower income of less than or equal to ₹1,00,000 and whose landholding was less than or equal to 2 acres were found to be able to apply for 6 to 7

different government scheme eligibility programs based upon the attributes of their profile. Other farmers would only qualify for fewer eligible programs based on their respective profiles. Final outputs generated by the system included recommendations on crops to grow, schemes that they were eligible to apply for, responses from the Chatbot and downloadable PDF certificates with their name on it.

The system provided high levels of accuracy, efficiency and usability to farmers in order to provide them with a comprehensive decision support system. The use of machine learning models in combination with rule-based and multilingual methodologies allows the proposed solution to successfully address all the limitations associated with the way in which eligibility is determined by existing methods. The findings support the conclusion that the Farmer Portal is a reliable, scalable and functional tool that will have a positive impact on agricultural production and improve the knowledge that farmers have concerning government programs.

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